Physics Informed Diffusion Models

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Outline

1. Recap Diffusion Models
2. Setup: Physics Informed Diffusion Models
3. Training Algorithm
4. Experiments
5. Discussion

Recap Diffusion Models

- First point
- Second point
- Third point

Physics Informed Diffusion Models

Definition

We assume our physics is described by a PDE on $\Omega \subseteq \mathbb{R}^d$ of the form

$$\mathcal{F}\left(u(\xi)\right) = 0 \quad \text{with} \quad \xi \in \Omega.$$

and we have a boundary operator

$$\mathcal{B}\left(u(\xi)\right) = 0 \quad \text{with} \quad \xi \in \partial\Omega.$$

where $u : \Omega \to \mathbb{R}^c$ is the solution of interest. To simplify, we write

$$R(u(\xi)) = [\mathcal{F}(u(\xi)), \mathcal{B}(u(\xi))]^\top$$

.

They only consider

$$[1, ..., n] \times [1, ..., n] \subseteq \mathbb{Z}_n \times \mathbb{Z}_n.$$

Similar to standard image based diffusion models, data is then considered to be given as $x_0 \in \mathbb{R}^{c \times n \times n}$ and $\mathcal{F}, \mathcal{B}$ are discretized accordingly.

Setting up Physics based Training Objective

The residual $\mathcal{R} = 0$ in the non discrete gives us an infinite amount of information. To utilize this, we introduce a loss based on $\mathcal{R}$ via a *virtual observable*.

$$q_{\mathcal{R}}(\cdot \mid x_0) \sim \mathcal{N}(\mathcal{R}(x_0), \sigma^2 I_d)$$

Definition

Define

$$p_\theta(\hat{r}) = \int_\Omega q_{\mathcal{R}}(\hat{r} \mid x_0) p_\theta(x_0) dx_0$$

then we can optimize

$$\arg\max_\theta \mathbb{E}_{\hat{r} \sim \delta_0} \left[\log p_\theta(\hat{r})\right] = \arg\max_\theta \mathbb{E}_{x_0 \sim p_\theta} \left[\log q_{\mathcal{R}}\left(\hat{r} = 0 \mid x_0\right)\right].$$

So we can train our model by evaluating x_0 on the the virtual observable log likelihood. Note that this loss is equivalent to

$$\arg\max_\theta \log p_\theta(0) = \arg\max_\theta \mathbb{E}_{x_0 \sim p_\theta} \left[\log q_{\mathcal{R}}\left(0 \mid x_0\right)\right]$$

We maintain the probabilistic perspective of generative models and introduce the residuals as *virtual observables* (Rixner & Koutsourelakis, 2021) $\hat{\boldsymbol{r}} = \mathbf{0}$, which we consider to be sampled from the distribution

$$q_{\mathcal{R}}(\hat{\boldsymbol{r}}|\boldsymbol{x}_0) = \mathcal{N}(\hat{\boldsymbol{r}}; \boldsymbol{\mathcal{R}}(\boldsymbol{x}_0), \sigma^2 \boldsymbol{I}). \tag{8}$$

$q_R = p_\theta$?

In the limit $\sigma^2 \to 0$, this recovers the deterministic setting of strictly enforcing the residual equations, as all probability mass is concentrated in $\boldsymbol{\mathcal{R}}(\boldsymbol{x}_0)$. This can be used to compute the virtual likelihood $p_\theta(\hat{\boldsymbol{r}})$, which we expand in terms of drawn samples $\boldsymbol{x}_0$ as

Makes no

$$p_\theta(\hat{\boldsymbol{r}}) = \int p_\theta(\hat{\boldsymbol{r}}, \boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_0 = \int q_{\mathcal{R}}(\hat{\boldsymbol{r}}|\boldsymbol{x}_0) p_\theta(\boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_0 = \mathbb{E}_{\boldsymbol{x}_0 \sim p_\theta(\boldsymbol{x}_0)} q_{\mathcal{R}}(\hat{\boldsymbol{r}}|\boldsymbol{x}_0). \tag{9}$$

sense

This factorization of $p_\theta(\hat{\boldsymbol{r}}, \boldsymbol{x}_0)$ is reasonable as the residual follows $\boldsymbol{x}_0$ via equation 8. The goal is to maximize the usual log-likelihood over the virtual observable $\hat{\boldsymbol{r}}$, i.e.,

$$\arg\max_\theta \mathbb{E}_{\hat{\boldsymbol{r}}} \left[\log p_\theta(\hat{\boldsymbol{r}})\right] = \arg\max_\theta \mathbb{E}_{\boldsymbol{x}_0 \sim p_\theta(\boldsymbol{x}_0)} \left[\log q_{\mathcal{R}}(\hat{\boldsymbol{r}} = \mathbf{0}|\boldsymbol{x}_0)\right]. \tag{10}$$

define $p_\theta(\hat{r})_\theta = \int_\Omega q_R(\hat{r}|x_0) p_\theta(x_0)\, dx_0$?

Figure

Introducing the Data Loss

Full Training Algorithm

Discussion

- They only consider 2d data and PDEs on $\mathbb{R}^2$, how this works on higher dimensions is not tested.